When the Leaderboard Flips

Non-technical report for decision-makers | Module 24, UC Berkeley Executive Education Post Graduate Program in AI and Machine Learning

Executive summary

This project asks whether an AI model that appears best on a standard test remains best when a user asks the same question in a different way. We examined four large language models across two reasoning benchmarks and five prompt formulations. The goal is to inform practical model selection, not to announce a universal winning model.

The decision problem

Organizations often compare AI models using a single accuracy score. Real users, however, phrase instructions differently, add irrelevant details, and change formatting. A small change in a model's score may affect operational decisions, especially when the leading models are closely matched.

What we studied

The analysis includes 100 distinct questions drawn from ARC-Challenge and GSM8K, arranged in two blocks. Each question was evaluated under five prompt conditions for four models, yielding 2,000 intended responses. Missing API acquisitions, unparseable answers, and wrong answers were tracked separately.

What we found

All four systems performed strongly on this limited set of questions, particularly in Block A. Some differences emerged across models, blocks and prompt formulations. Because results are near the accuracy ceiling and the question sample is small, the observed ordering cannot be treated as a definitive general-purpose leaderboard.

Why response completeness matters

A response that never arrives or cannot be interpreted is not equivalent to a successful response. An operational evaluation should report the full end-to-end result, not just accuracy among answers that were successfully parsed.

Business and operational relevance

A deployment team can use the framework to compare candidates on its own workload, inspect the prompt conditions that matter most, and distinguish apparent ranking changes from stable

differences. It should not make a procurement or safety decision solely from this preliminary benchmark.

Limitations

Only two benchmark families and 100 distinct questions were analyzed. Questions were repeatedly measured under multiple prompt conditions, so 2,000 responses do not constitute 2,000 independent questions. The study does not measure cost, service-level reliability across time, or deployment performance for a specific organization.

Recommendations and next steps

For an initial pilot, collect representative tasks and prompt variants, record failures, compare models on matched questions, and validate decisions on held-out questions. A subsequent journal-scale study can include more difficult benchmarks, more models and stronger robustness-aware selection policies.

Deliverables

The accompanying GitHub repository includes one main executable notebook, the frozen analytical dataset, generated figure files, technical report, figure caption notes and a reproducibility manifest.